MScT Complex Systems Engineering

Centrale Méditerranée — 2026–2027

Objectives

The goal of this project is to apply the knowledge acquired during the course to investigate a specific question or problem using data and Python.

You will conduct an independent data science project in which you collect, understand, process, analyze, and interpret one or more datasets. Your objective is to produce meaningful findings supported by data, rather than simply explore a dataset or demonstrate your ability to write code.

You are expected to follow a modern Python workflow using `uv` for project and environment management, **Git** for version control, and **Pandas** for data processing.

Above all, I want you to investigate a subject that genuinely interests you and discover something worth sharing. A successful project combines a well-defined question, appropriate methods, critical thinking, and an original perspective.

1. Project requirements

Your submission must include at least one Jupyter notebook (`.ipynb`) presenting your investigation in a clear, structured, and reproducible way.

The notebook must contain the following sections.

1.1. Introduction

Introduce your topic and explain why it is interesting or relevant. State the question or problem you intend to investigate and briefly describe your approach.

Include a URL to each dataset used and explain how the data will help you investigate your question.

Your project must involve quantitative data at some point. The data may be quantitative in their original form or become quantitative through appropriate processing. Tabular data formats such as CSV and Excel are often convenient, but other formats are acceptable when appropriate.

1.2. Dataset description

Describe the source and characteristics of each dataset.

Explain, where this information is available:

- Who created or published the data.
- Why and how the data were collected.
- What the observations and variables represent.
- The population, geographical area, and period covered.
- Any relevant sampling procedures, biases, limitations, or missing information.
- The format, size, and structure of the data.

Distinguish the original data from any data derived or created during your analysis. Cite your sources and provide enough information for another person to identify and obtain the same data.

1.3. Research question and hypotheses

Formulate a clear, specific research question that can reasonably be investigated using your data.

State one or more hypotheses when appropriate. Explain what you expect to find and why. Your analysis should provide evidence that supports, contradicts, or qualifies these hypotheses.

1.4. Descriptive statistics and data quality

Examine the structure and quality of your dataset before conducting the main analysis.

Depending on the data, this may include:

- Number of observations and variables.
- Data types and variable descriptions.
- Missing values, duplicates, and inconsistent records.
- Distributions of important variables.
- Mean, median, standard deviation, quantiles, or other relevant statistics.
- Outliers, potential measurement errors, and data-quality issues.

Choose indicators that are appropriate for your data and explain what they reveal. Do not include statistics merely to fill a section.

1.5. Data processing and analysis

Clean and transform your data as necessary, then conduct an analysis designed to answer your research question.

Depending on your project, this may include:

- Filtering, joining, aggregating, or reshaping datasets.
- Handling missing values, duplicates, and inconsistent data.
- Creating new variables or deriving quantitative indicators.
- Comparing groups, studying relationships, or investigating changes over time.
- Creating relevant visualizations, including maps when appropriate.
- Applying statistical methods or building predictive models when they genuinely contribute to answering the question.

Explain the reasoning behind your choices. Accompany important tables, figures, and quantitative results with an interpretation.

Your analysis must be focused and purposeful. Avoid long, unstructured exploratory sessions, redundant visualizations, and calculations unrelated to your research question.

You are not expected to use every technique covered in the course. Select the methods that best suit your problem and the available data.

1.6. Results and discussion

Present your main findings and explain their significance.

Discuss the extent to which your results answer the research question and support or contradict your hypotheses. Distinguish observations from interpretations and acknowledge uncertainty where relevant.

Be careful when interpreting relationships between variables: correlation alone does not establish causation.

Discuss the limitations of your data and methods, including potential biases, confounding factors, and alternative explanations where relevant. Explain what additional data or further investigations could improve your conclusions.

1.7. Conclusion

Summarize the most important findings and answer your initial research question as directly as possible.

Explain what you learned from the investigation and identify any remaining questions or promising directions for future work.

The conclusion should synthesize your results rather than simply repeat the introduction or list the steps you performed.

2. Scientific quality and originality

This project is an opportunity to investigate a subject that interests you and to develop your own perspective on it.

Look for an unexpected relationship, investigate a surprising phenomenon, challenge an assumption, compare competing explanations, or reveal something that is not immediately obvious from the data.

Originality does not necessarily mean discovering something that nobody has ever studied before. It can also mean asking an interesting question that has not been addressed in your particular context, combining datasets in a useful way, or presenting an established phenomenon from a new perspective.

A good project does not need sophisticated machine learning or complicated mathematics. A simple, carefully designed analysis that produces a convincing and interesting result is preferable to a technically complex project without a clear purpose.

Your objective is to make me interested in what you have discovered. I want to understand your question, follow your reasoning, examine your evidence, and learn something from your work.

3. Use of AI tools

You may use AI tools to assist you with programming, debugging, understanding documentation, exploring possible methods, or improving the presentation of your work. However, these tools must support your investigation, not replace your thinking.

I do not want to read pages of generic, AI-generated text that could have been written about any dataset or any subject. I want to read your own reasoning and discover something interesting through your research.

In particular:

- Write explanations that reflect your actual observations, decisions, and conclusions.
- Do not include generic introductions, lengthy descriptions, or repetitive interpretations simply to make your notebook look more substantial.
- Do not ask an AI tool to invent findings, references, explanations, or conclusions that you have not verified.
- Verify AI-generated code, check its outputs, and make sure you understand the methods you use.
- If you use AI to help formulate hypotheses or select methods, evaluate these suggestions critically rather than accepting them automatically.

You remain responsible for the accuracy, originality, and quality of your submission, including any code or text produced with AI assistance.

Quality matters more than quantity. A concise notebook with a compelling research question, a few well-chosen analyses, and insightful interpretations is much better than a long notebook full of generic text and redundant plots.

4. Good practices and technical requirements

4.1. Project structure and version control

Store your notebook and all other relevant project files in a Git repository. The repository should contain, as appropriate:

- At least one `.ipynb` notebook.
- The original dataset(s), when their size and licensing permit.
- Any additional files required to run or understand the analysis.
- `pyproject.toml`, defining the project and its dependencies.
- `uv.lock`, recording the resolved dependency versions.
- `.gitignore`, excluding unnecessary files and sensitive information.
- `README.md`, explaining the project and how to run it.

You must use `uv` to manage your Python project, dependencies, and virtual environment.

You must use **Git** for version control. Your repository must be publicly accessible on GitHub or GitLab (or any other public Git hosting service).

4.2. README and reproducibility

Your `README.md` must include:

- The project title and a concise description of the research question.
- The main findings or a summary of what the project investigates.

- The data sources, with links to the original datasets.
- Instructions for installing the dependencies and running the project using `uv`.
- Any additional steps required to obtain or prepare the data.
- **The GIT URL of your public repository.**

Your project must be reproducible by another person following these instructions in a fresh environment.

Before submitting, restart the notebook kernel and execute all cells from beginning to end. Make sure the notebook runs without errors and that the outputs correspond to the code.

If a dataset is too large to include in your repository or submission, document how to obtain the exact version used. If access requires credentials or an API key, explain the necessary setup without publishing any secrets. Also if you use an API key or token, please provide one that I can use to reproduce your results.

I will not create an account or pay for a service to access your data.

4.3. Notebook presentation

Use Markdown to structure your notebook. Include appropriate section headings, concise explanations, and clear transitions between the research question, methodology, results, and conclusions.

Use meaningful variable names, readable code, and appropriately labelled figures. Include units, legends, and informative titles when relevant.

5. Methodology and data requirements

The following requirements apply to every project.

- **A distinct main dataset:** Each student must work with a different main dataset or a genuinely different combination of datasets. Coordinate with your classmates to avoid duplicate projects.
- **Secondary datasets:** You may use datasets selected by other students as secondary sources when combining them serves a clear analytical purpose, for example to enrich your main dataset or investigate relationships between different phenomena.
- **Original data:** Start from the raw data as obtained from the original source. Do not manually edit the original files in Excel or other software before analysis. All data cleaning and processing must be performed in Python.
- **Quantitative analysis:** Your project must include quantitative data and meaningful quantitative analysis.

Expectations

Research question and originality — 3 points

The question is clear, relevant, and suitable for investigation using data. The project demonstrates curiosity, initiative, and an effort to produce interesting findings.

Data understanding and preparation — 4 points

The student understands the data sources, identifies relevant data-quality issues, and performs appropriate cleaning and transformations in Python.

Analysis and interpretation — 5 points

The methods are appropriate to the question. Quantitative results and visualizations are meaningful, correctly interpreted, and used to support evidence-based conclusions.

Python implementation — 3 points

The code is correct, readable, and appropriately organized. Pandas is used for data processing, and the implementation is suitable for the task.

Reproducibility and documentation — 3 points

The project uses `uv` and Git appropriately, includes the required files, and provides clear instructions that allow another person to reproduce the analysis.

Communication and presentation — 2 points

The notebook is well structured, clearly written, and professionally presented. Figures, tables, and explanations are easy to understand, with few or no spelling or formatting mistakes.

A project is not rewarded simply for using advanced techniques, producing a large number of charts, or including a great deal of text. The quality of the reasoning and the relevance of the findings matter more than the volume or complexity of the work.

7. Submission format

Submit your project through the Centrale Méditerranée Moodle platform.

Your submission must be a compressed archive (ZIP, RAR, or equivalent) containing all the files.

The archive and public Git repository must describe the same final version of the project.

If a dataset is too large to upload to Moodle, provide a reliable way to download the exact version of the original data and document the procedure in your README.md.

If you cannot submit your work through Moodle, contact me at **virgile.pesce@univ-amu.fr**.

8. Tips and resources

Finding datasets

You can find datasets on many public platforms, including:

- data.gouv.fr — French public-sector data.
- Data Sud — regional data for southern France.
- Nos Données — open-data resources.
- Kaggle — datasets covering a wide range of subjects.
- GitHub — datasets and data-related projects.
- Official research institutions and scientific repositories — research datasets and associated documentation.

9. Questions and difficulties

If you have questions or encounter difficulties, feel free to contact me at **virgile.pesce@univ-amu.fr**.

Do not wait until the last minute to start this project.

Good luck, and I look forward to seeing your investigations!